Lab Assignment-17.1

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Batch: 14

Task1 – Customer Data Cleaning

Task:

Use AI to generate a Python script for cleaning a customer dataset.

PROMPT: Generate a Python script using Pandas to clean a customer dataset loaded from a CSV file named "shopping_trends.csv". The script should handle missing values in columns such as age, income, and city by filling or removing them appropriately. It should remove duplicate records from the dataset. Convert all text values in the city column to lowercase to ensure consistency. Encode categorical variables like gender and city into numerical format using suitable encoding techniques. Finally, display the cleaned dataset and ensure it is ready for further data analysis.

```
[1]
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load the dataset
try:
    df = pd.read_csv('shopping_trends.csv')
    print("Dataset loaded successfully.")
except FileNotFoundError:
    print("Error: 'shopping_trends.csv' not found. Please make sure the file is in the correct directory.")
    df = pd.DataFrame() # Create an empty DataFrame to avoid errors in subsequent steps

# Display initial info and head to understand the data
if not df.empty:
    print("\nInitial dataset info:")
    df.info()
    print("\nFirst 5 rows of the original dataset:")
    display(df.head())

Error: 'shopping_trends.csv' not found. Please make sure the file is in the correct directory.

[2]
if not df.empty:
    print("\nMissing values before handling:")
    print(df[['Age', 'Income', 'City']].isnull().sum())

    # Remove rows with missing values in 'Age', 'Income', or 'City'
    initial_rows = df.shape[0]
    df.dropna(subset=['Age', 'Income', 'City'], inplace=True)
    rows_after_na = df.shape[0]
    print(f"\nRemoved {initial_rows - rows_after_na} rows with missing values in Age, Income, or City.")

    print("\nMissing values after handling:")
    print(df[['Age', 'Income', 'City']].isnull().sum())

[3]
if not df.empty:
    initial_rows = df.shape[0]
    df.drop_duplicates(inplace=True)
    rows_after_duplicates = df.shape[0]
    print(f"\nRemoved {initial_rows - rows_after_duplicates} duplicate rows.")
    print(f"Dataset now has {df.shape[0]} rows.")

[4]
if not df.empty and 'City' in df.columns:
    df['City'] = df['City'].str.lower()
    print("\n'City' column converted to lowercase.")
    display(df[['City']].head())
```

```python
if not df.empty:
    initial_rows = df.shape[0]
    df.drop_duplicates(inplace=True)
    rows_after_duplicates = df.shape[0]
    print(f"\nRemoved {initial_rows - rows_after_duplicates} duplicate rows.")
    print(f"Dataset now has {df.shape[0]} rows.")
```

```python
if not df.empty and 'City' in df.columns:
    df['City'] = df['City'].str.lower()
    print("\n'City' column converted to lowercase.")
    display(df[['City']].head())
```

```python
if not df.empty:
    # Encode 'Gender'
    if 'Gender' in df.columns:
        gender_mapping = {'Male': 0, 'Female': 1}
        df['Gender_Encoded'] = df['Gender'].map(gender_mapping)
        print("\n'Gender' column encoded.")
        print("Original Gender unique values:", df['Gender'].unique())
        print("Encoded Gender unique values:", df['Gender_Encoded'].unique())

    # Encode 'City'
    if 'City' in df.columns:
        le = LabelEncoder()
        df['City_Encoded'] = le.fit_transform(df['City'])
        print("\n'City' column encoded.")
        print("Original City unique values (first 5):", df['City'].unique()[:5])
        print("Encoded City unique values (first 5):", df['City_Encoded'].unique()[:5])

    print("\nCleaned dataset info:")
    df.info()
    print("\nFirst 5 rows of the cleaned dataset:")
    display(df.head())
```

The dataset has been cleaned and is now ready for further analysis!

```python
!unzip -o '/archive (8).zip' -d '.'
```

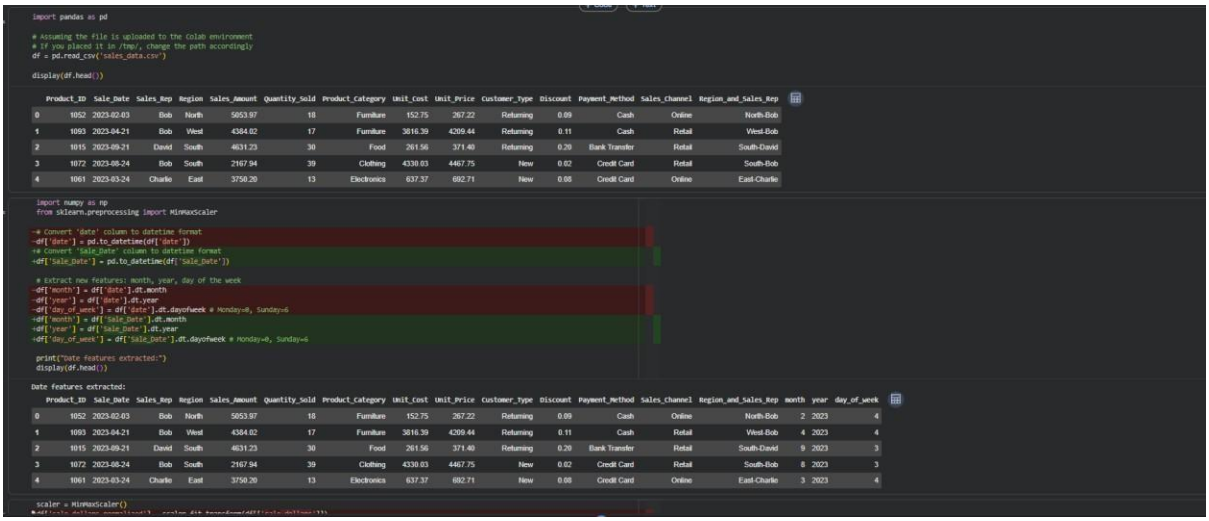
```
Archive:  /archive (8).zip
  inflating: ./shopping_trends.csv
  inflating: ./shopping_trends_updated.csv
```

Task2 – Sales DataPreprocessing

Task:

Preprocess sales transaction data using AI assistance.

PROMPT: Generate a Python script using Pandas and NumPy to preprocess a sales dataset loaded from a CSV file named "sales_data.csv". Convert the date column into datetime format and extract new features such as month, year, and day of the week. Normalize the sales amount column using an appropriate scaling method like Min-Max normalization. Detect outliers in the dataset using techniques such as the IQR method or Z-score and handle them by removing or capping extreme values. Finally, display the transformed dataset with the new features, ensuring it is ready for further analysis



```
import pandas as pd

# Assuming the file is uploaded to the Colab environment
# If you placed it in /tmp/, change the path accordingly
df = pd.read_csv('sales_data.csv')

display(df.head())
```

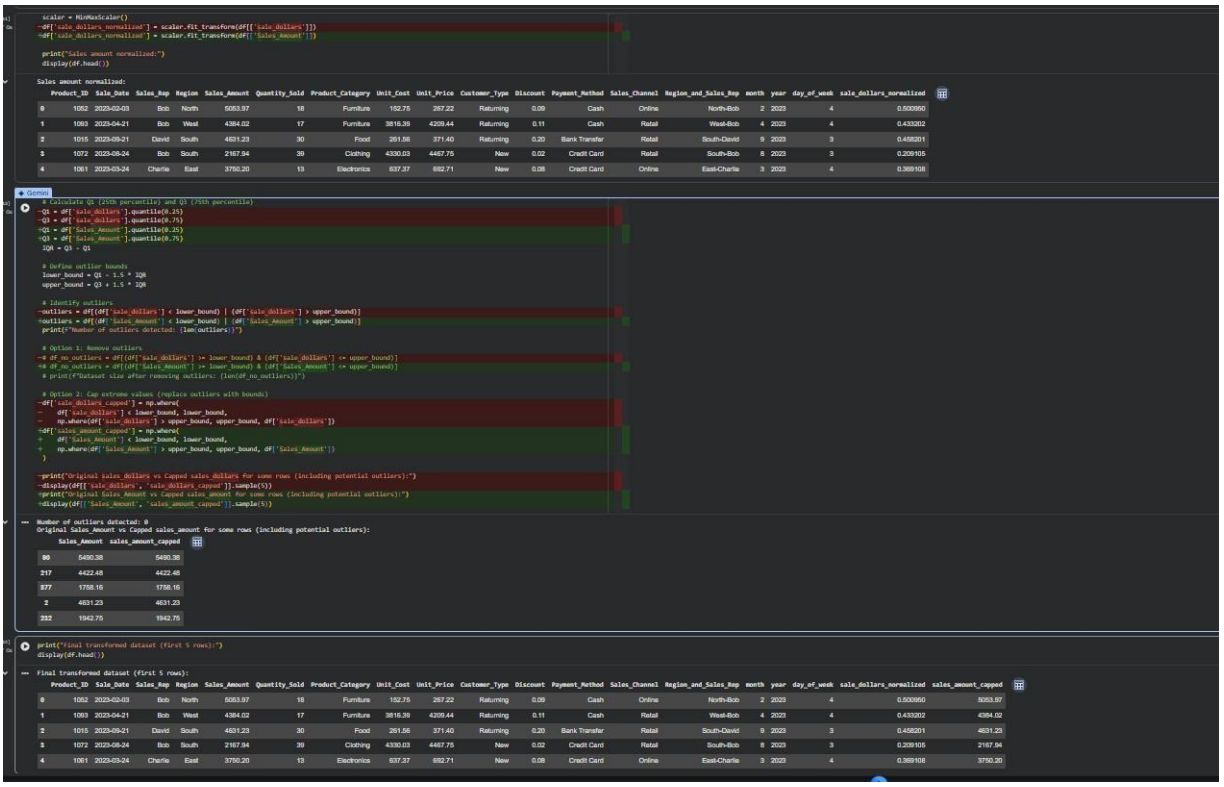
	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie

```
import numpy as np
from sklearn.preprocessing import MinMaxScaler

-# Convert 'date' column to datetime format
-df['date'] = pd.to_datetime(df['date'])
+# Convert 'Sale_Date' column to datetime format
+df['Sale_Date'] = pd.to_datetime(df['Sale_Date'])

# Extract new features: month, year, day of the week
-df['month'] = df['date'].dt.month
-df['year'] = df['date'].dt.year
-df['day_of_week'] = df['date'].dt.dayofweek # Monday=0, Sunday=6
+df['month'] = df['Sale_Date'].dt.month
+df['year'] = df['Sale_Date'].dt.year
+df['day_of_week'] = df['Sale_Date'].dt.dayofweek # Monday=0, Sunday=6

print("Date features extracted:")
display(df.head())
```

Date features extracted:

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	month	year	day_of_week
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4

```
scaler = MinMaxScaler()
```



```
scaler = MinMaxScaler()
-df['sale_dollars_normalized'] = scaler.fit_transform(df[['sale_dollars']])
+df['sale_dollars_normalized'] = scaler.fit_transform(df[['Sales_Amount']])

print("Sales amount normalized:")
display(df.head())
```

Sales amount normalized:

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	month	year	day_of_week	sale_dollars_normalized
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4	0.500950
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4	0.433202
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3	0.458201
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3	0.209105
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4	0.369108

Gemini

```
# Calculate Q1 (25th percentile) and Q3 (75th percentile)
-Q1 = df['sale_dollars'].quantile(0.25)
-Q3 = df['sale_dollars'].quantile(0.75)
+Q1 = df['Sales_Amount'].quantile(0.25)
+Q3 = df['Sales_Amount'].quantile(0.75)
IQR = Q3 - Q1

# Define outlier bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Identify outliers
-outliers = df[(df['sale_dollars'] < lower_bound) | (df['sale_dollars'] > upper_bound)]
+outliers = df[(df['Sales_Amount'] < lower_bound) | (df['Sales_Amount'] > upper_bound)]
print(f"Number of outliers detected: {len(outliers)}")

# Option 1: Remove outliers
-# df_no_outliers = df[(df['sale_dollars'] >= lower_bound) & (df['sale_dollars'] <= upper_bound)]
+# df_no_outliers = df[(df['Sales_Amount'] >= lower_bound) & (df['Sales_Amount'] <= upper_bound)]
# print(f"Dataset size after removing outliers: {len(df_no_outliers)}")

# Option 2: Cap extreme values (replace outliers with bounds)
-df['sale_dollars_capped'] = np.where(
-    df['sale_dollars'] < lower_bound, lower_bound,
-    np.where(df['sale_dollars'] > upper_bound, upper_bound, df['sale_dollars'])
-df['sales_amount_capped'] = np.where(
+    df['Sales_Amount'] < lower_bound, lower_bound,
+    np.where(df['Sales_Amount'] > upper_bound, upper_bound, df['Sales_Amount'])
 )

-print("Original sale_dollars vs Capped sale_dollars for some rows (including potential outliers):")
-display(df[['sale_dollars', 'sale_dollars_capped']].sample(5))
+print("Original Sales_Amount vs Capped sales_amount for some rows (including potential outliers):")
+display(df[['Sales_Amount', 'sales_amount_capped']].sample(5))
```

```
Number of outliers detected: 0
Original Sales_Amount vs Capped sales_amount for some rows (including potential outliers):
```

	Sales_Amount	sales_amount_capped
90	5490.38	5490.38
217	4422.48	4422.48
377	1758.16	1758.16
2	4631.23	4631.23
222	1942.75	1942.75

```
print("Final transformed dataset (first 5 rows):")
display(df.head())
```

Final transformed dataset (first 5 rows):

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	month	year	day_of_week	sale_dollars_normalized	sales_amount_capped
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4	0.500950	5053.97
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4	0.433202	4384.02
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3	0.458201	4631.23
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3	0.209105	2167.94
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4	0.369108	3750.20

Task3– HealthcareData Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

PROMPT: Generate a Python script using Pandas and Scikit-learn to clean and preprocess a healthcare dataset loaded from a CSV file. The script should handle missing values in columns such as blood_pressure and cholesterol by filling or removing them appropriately. Encode categorical features using techniques like label encoding or one-hot encoding. Scale numerical features using methods such as Min-Max scaling or

Standard scaling. Finally, split the dataset into training and testing sets using an appropriate ratio (e.g., 80:20) and display the processed data, ensuring it is ready for machine learning models.

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Load the dataset (replace 'healthcare_data.csv' with your actual file name)
try:
    df = pd.read_csv('healthcare_data.csv')
    print("Dataset loaded successfully. First 5 rows:")
    display(df.head())
except FileNotFoundError:
    print("Error: 'healthcare_data.csv' not found. Please upload the file to your Colab environment or specify the correct path.")
    # Create a dummy DataFrame for demonstration if file not found
    data = {
        'Patient_ID': range(1, 11),
        'Age': [30, 45, 60, 25, 50, 35, 70, 40, 55, 65],
        'Gender': ['Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female'],
        'Blood_Pressure': [120, 140, None, 110, 130, 125, 150, 118, None, 135],
        'Cholesterol': [180, 220, 200, 170, None, 190, 240, 160, 210, None],
        'Smoker': ['Yes', 'No', 'No', 'Yes', 'No', 'No', 'Yes', 'No', 'Yes', 'No'],
        'Diagnosis': ['Healthy', 'High BP', 'High Chol', 'Healthy', 'High BP', 'Healthy', 'High BP', 'Healthy', 'High Chol', 'High BP']
    }
    df = pd.DataFrame(data)
    print("Using dummy data for demonstration.")
    display(df.head())
```

```
Error: 'healthcare_data.csv' not found. Please upload the file to your Colab environment or specify the correct path.
Using dummy data for demonstration.
```

	Patient_ID	Age	Gender	Blood_Pressure	Cholesterol	Smoker	Diagnosis
0	1	30	Male	120.0	180.0	Yes	Healthy
1	2	45	Female	140.0	220.0	No	High BP
2	3	60	Male	NaN	200.0	No	High Chol
3	4	25	Female	110.0	170.0	Yes	Healthy
4	5	50	Male	130.0	NaN	No	High BP

```python
# Assuming 'Diagnosis' is the target variable
target_column = 'Diagnosis'

# Separate features (X) and target (y)
X = df.drop(columns=[target_column])
y = df[target_column]

# Identify numerical and categorical columns for preprocessing
numerical_cols = X.select_dtypes(include=np.number).columns.tolist()
categorical_cols = X.select_dtypes(include='object').columns.tolist()

print(f"Numerical columns: {numerical_cols}")
print(f"Categorical columns: {categorical_cols}")
```

```
Numerical columns: ['Patient_ID', 'Age', 'Blood_Pressure', 'Cholesterol']
Categorical columns: ['Gender', 'Smoker']
```

```python
# Create preprocessing pipelines for numerical and categorical features
numerical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='mean')),  # Fill missing numerical values with the mean
    ('scaler', StandardScaler())                  # Scale numerical features
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')), # Fill missing categorical values with the most frequent
    ('onehot', OneHotEncoder(handle_unknown='ignore'))   # One-hot encode categorical features
])

# Combine preprocessing steps using ColumnTransformer
# This applies different transformers to different columns
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_cols),
        ('cat', categorical_transformer, categorical_cols)
    ])

# Create a full preprocessing pipeline
full_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor)
])

# Apply the preprocessing to the features
X_processed = full_pipeline.fit_transform(X)

# Convert processed features back to DataFrame (optional, but good for inspection)
# Get feature names after one-hot encoding
# Need to access the 'onehot' step within the 'cat' transformer of the preprocessor
encoded_feature_names = full_pipeline.named_steps['preprocessor'].named_transformers_['cat'].named_steps['onehot'].get_feature_names_out(categorical_cols)
new_feature_names = numerical_cols + encoded_feature_names.tolist()

X_processed_df = pd.DataFrame(X_processed, columns=new_feature_names)

print("Processed features (first 5 rows):")
display(X_processed_df.head())
```

```
Processed features (first 5 rows):
```

	Patient_ID	Age	Blood_Pressure	Cholesterol	Gender_Female	Gender_Male	Smoker_No	Smoker_Yes
0	-1.566699	-1.218544	-0.783818	-0.727632	0.0	1.0	0.0	1.0
1	-1.218544	-0.174078	1.060460	1.063462	1.0	0.0	1.0	0.0
2	-0.870388	0.870388	0.000000	0.167915	0.0	1.0	1.0	0.0
3	-0.522233	-1.566699	-1.705957	-1.175406	1.0	0.0	0.0	1.0
4	-0.174078	0.174078	0.138321	0.000000	0.0	1.0	1.0	0.0

```python
# Split the processed data into training and testing sets (80:20 ratio)
X_train, X_test, y_train, y_test = train_test_split(
    X_processed_df, y, test_size=0.3, random_state=42, stratify=y # stratify helps maintain class distribution
)

print(f"Shape of X_train: {X_train.shape}")
print(f"Shape of X_test: {X_test.shape}")
print(f"Shape of y_train: {y_train.shape}")
print(f"Shape of y_test: {y_test.shape}")

print("\nFirst 5 rows of X_train:")
display(X_train.head())

print("\nFirst 5 rows of y_train:")
display(y_train.head())
```

```
Shape of X_train: (7, 8)
Shape of X_test: (3, 8)
Shape of y_train: (7,)
Shape of y_test: (3,)

First 5 rows of X_train:
```

	Patient_ID	Age	Blood_Pressure	Cholesterol	Gender_Female	Gender_Male	Smoker_No	Smoker_Yes
4	-0.174078	0.174078	0.138321	0.000000	0.0	1.0	1.0	0.0
8	1.218544	0.522233	0.000000	0.615689	0.0	1.0	0.0	1.0
3	-0.522233	-1.566699	-1.705957	-1.175406	1.0	0.0	0.0	1.0
9	1.566699	1.218544	0.599390	0.000000	1.0	0.0	1.0	0.0
1	-1.218544	-0.174078	1.060460	1.063462	1.0	0.0	1.0	0.0

```
First 5 rows of y_train:
```

	Diagnosis
4	High BP
8	High Chol
3	Healthy
9	High BP
1	High BP

```
dtype: object
```

Task 4 – Employee Salary Data Preprocessing Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions

PROMPT: Generate a Python script using Pandas and Scikit-learn to clean and preprocess an employee salary dataset loaded from a CSV file. The script should handle missing values by filling or removing them appropriately. Detect and remove salary outliers using techniquessuch as the IQR method or Z-score. Standardize text values in the department column by converting them to lowercase for consistency. Apply label encoding to categorical columns such as job location and department. Finally, display the cleaned and structured dataset, ensuring it is ready for HR analytics.

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder

# Load the dataset (replace 'Employee_Salary_Dataset.csv' with your actual file name)
try:
    df = pd.read_csv('Employee_Salary_Dataset.csv')
    print("Dataset loaded successfully. First 5 rows:")
    display(df.head())
except FileNotFoundError:
    print("Error: 'Employee_Salary_Dataset.csv' not found. Please upload the file or specify the correct path.")
    # Create a dummy DataFrame for demonstration if file not found
    data = {
        'Employee_ID': range(1, 11),
        'Age': [25, 30, 35, 40, 28, 45, 50, 32, 38, 29],
        'Gender': ['Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female'],
        'Department': ['Sales', 'Marketing', 'IT', 'HR', 'sales', 'Finance', 'IT', 'Marketing', 'HR', 'sales'],
        'Job_Title': ['Sales Rep', 'Marketing Analyst', 'Software Engineer', 'HR Manager', 'Sales Manager', 'Financial Analyst', 'Data Scientist', 'Marketing Specialist', 'HR Coordinator', 'Sales Associate'],
        'Job_Location': ['New York', 'London', 'Remote', 'New York', 'London', 'Remote', 'New York', 'London', 'Remote', 'New York'],
        'Years_Of_Experience': [2, 5, 8, 12, 4, 15, 10, 6, 9, 3],
        'Salary': [50000, 60000, 80000, np.nan, 70000, 150000, 90000, 65000, 75000, 55000] # Added a potential outlier at 150k and a NaN
    }
    df = pd.DataFrame(data)
    print("Using dummy data for demonstration.")
    display(df.head())

# Display initial info
print("\nInitial DataFrame Info:")
df.info()
```

```
Dataset loaded successfully. First 5 rows:
```

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000

```
Initial DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 5 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                35 non-null     int64
 1   Experience_Years  35 non-null     int64
 2   Age               35 non-null     int64
 3   Gender            35 non-null     object
 4   Salary            35 non-null     int64
dtypes: int64(4), object(1)
memory usage: 1.5+ KB
```

```python
# Fill missing 'Salary' values with the median
if 'Salary' in df.columns:
    median_salary = df['Salary'].median()
    df['Salary'].fillna(median_salary, inplace=True)
    print(f"Filled missing 'Salary' values with median: {median_salary}")

# For simplicity, if any other crucial columns have NaNs, we'll drop those rows.
# This might need to be adjusted based on the specific dataset and domain knowledge.
df.dropna(inplace=True)

print("\nDataFrame after handling missing values:")
df.info()
```

```
Filled missing 'Salary' values with median: 250000.0

DataFrame after handling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 5 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                35 non-null     int64
 1   Experience_Years  35 non-null     int64
 2   Age               35 non-null     int64
 3   Gender            35 non-null     object
 4   Salary            35 non-null     int64
dtypes: int64(4), object(1)
memory usage: 1.5+ KB
/tmp/ipykernel_355/3065862150.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df['Salary'].fillna(median_salary, inplace=True)
```

2. Detect and Remove Salary Outliers (IQR Method)

We'll use the Interquartile Range (IQR) method to identify and remove outliers in the 'Salary' column.

```python
if 'Salary' in df.columns:
    Q1 = df['Salary'].quantile(0.25)
    Q3 = df['Salary'].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Identify outliers
    outliers = df[(df['Salary'] < lower_bound) | (df['Salary'] > upper_bound)]
    print(f"\nNumber of salary outliers detected: {len(outliers)}")
    if not outliers.empty:
        print("Salary Outliers:")
        display(outliers)

    # Remove outliers
    df = df[~((df['Salary'] < lower_bound) | (df['Salary'] > upper_bound))]
    print("\nDataFrame after removing salary outliers:")
    display(df.head())
else:
    print("No 'Salary' column found for outlier detection.")
```

```
Number of salary outliers detected: 2
Salary Outliers:
```

	ID	Experience_Years	Age	Gender	Salary
27	28	27	62	Female	10000000
33	34	19	53	Female	9300000

```
DataFrame after removing salary outliers:
```

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000

3. Standardize Text Values in 'Department' Column

Convert all department names to lowercase for consistency.

```
if 'Department' in df.columns:
    df['Department'] = df['Department'].str.lower()
    print("\n'Department' column after standardization (lowercase):")
    print(df['Department'].value_counts())
else:
    print("No 'Department' column found for standardization.")
```

```
No 'Department' column found for standardization.
```

4. Apply Label Encoding to Categorical Columns

We'll encode 'Gender', 'Department', and 'Job_Location' using `LabelEncoder`.

```
categorical_cols_to_encode = ['Gender', 'Department', 'Job_Location']

for col in categorical_cols_to_encode:
    if col in df.columns:
        le = LabelEncoder()
        df[col + '_Encoded'] = le.fit_transform(df[col])
        print(f"\nLabel encoded '{col}':")
        display(df[[col, col + '_Encoded']].drop_duplicates().sort_values(by=col + '_Encoded'))
    else:
        print(f"Column '{col}' not found for encoding.")

print("\nFinal Cleaned and Preprocessed Dataset (first 5 rows):")
display(df.head())

print("\nFinal DataFrame Info:")
df.info()
```

```
Label encoded 'Gender':
```

	Gender	Gender_Encoded
0	Female	0
1	Male	1

```
Column 'Department' not found for encoding.
Column 'Job_Location' not found for encoding.

Final Cleaned and Preprocessed Dataset (first 5 rows):
```

	ID	Experience_Years	Age	Gender	Salary	Gender_Encoded
0	1	5	28	Female	250000	0
1	2	1	21	Male	50000	1
2	3	3	23	Female	170000	0
3	4	2	22	Male	25000	1
4	5	1	17	Male	10000	1

```
Final DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
Index: 33 entries, 0 to 34
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                33 non-null     int64
 1   Experience_Years  33 non-null     int64
 2   Age               33 non-null     int64
 3   Gender            33 non-null     object
 4   Salary            33 non-null     int64
 5   Gender_Encoded    33 non-null     int64
dtypes: int64(5), object(1)
memory usage: 1.8+ KB
```